

Autonomous AI Web Quality Engine :

Summary:

This project focuses on building an AI-driven autonomous engine that can crawl and analyze web applications without human intervention. The system handles authentication, explores website structure, and intelligently understands page types using DOM, visual, and content analysis. It automatically detects functional, UI, performance, accessibility, and API-related defects. All issues are classified, mapped in a knowledge graph, and used to generate a Hygiene Score that measures overall website quality. The final results are presented through a structured report and interactive dashboard.

Architecture Layout:

1. User Input Url :

- The user provides the web application URL as input

2. Authentication Handler :

Authentication Handler automates login Based on

- form base login : like username Password
- OAuth2 :Thirt Party Login (Google ,Microsoft)
- Token Based Authentication :JWT
- Session Management : Cookies, JWT persistence

3. Web Crawler :

The main purpose of a Web Crawler is to systematically explore a website, discover all reachable pages, understand the navigation structure, and collect page content and metadata for further analysis. The crawler collects page data, which is then processed by the analysis engine for defect detection.

4. Page Understanding AI Engine :

AI Engine analyses the page and interprets the page structure and behavior .

it have four works :

- DOM Structure Analysis : it means it will check how many input fields are there, how many button are there .
- Visual Analysis : it will check layout of structure .
- Content Analysis:it helps the AI engine understand page intent
- Behavioral Analysis: Button click test : it will check all the button are work properly or not

5. Bug Detection Engines :

Bug Detection Engines automatically identify functional, UI, performance, accessibility, and API-related defects through structured analysis and response monitoring.

6. Hygiene Classification :

The main purpose of Hygiene Classification is how serious the problem is.

EX: Broken Link is major, minor, critical

7. Defect Knowledge Graph :

Knowledge graph connects pages, elements & defects which page has more problematic, which category has more impact, which issue repeatedly happens.

8. Hygiene Score Calculation :

Hygiene Score is nothing but health of website

Hygiene Score measures overall website quality, improvement tracking

9. Final Report & Dashboard :

presentation layer Final Report & Dashboard Report - it will detect issues in structured format, also provide page-wise hygiene score & provide Overall Hygiene Score. Based on that report we can analyze our project which part improvement is required. And many more things.

Conclusion :

This architecture represents a fully autonomous AI-driven QA system that crawls, understands, detects, classifies, and scores web application defects without human intervention.